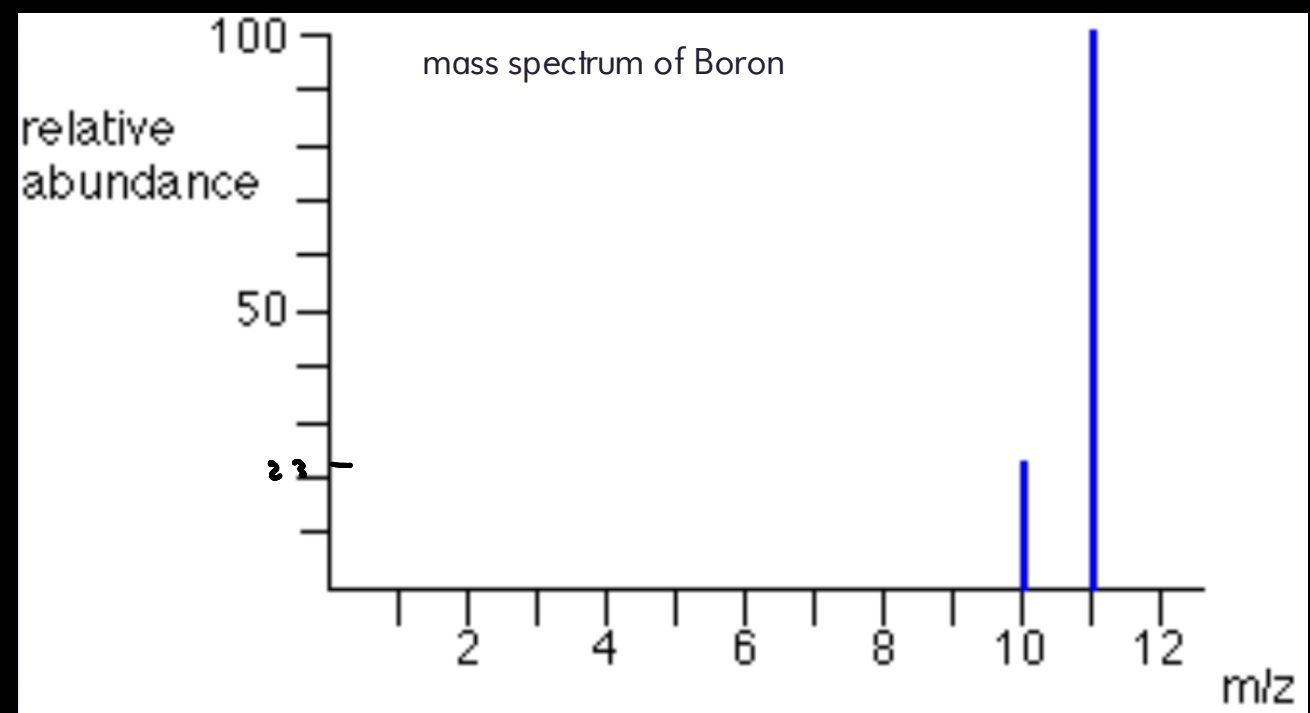


Analyzing Spectras

Spectra of monatomic Elements



$$R.A.M = \frac{\sum (\text{isotope mass} \times \text{isotope abundance})}{100}$$

For this example the abundance for :

$$B-10 \text{ is } \frac{23}{12} \times 100\% = 18.7\%$$

$$B-11 \text{ is } \frac{106}{128} \times 100\% = 81.3\%$$

$$R.A.M = \frac{(10 \times 18.7) + (11 \times 81.3)}{100} = 10.8$$

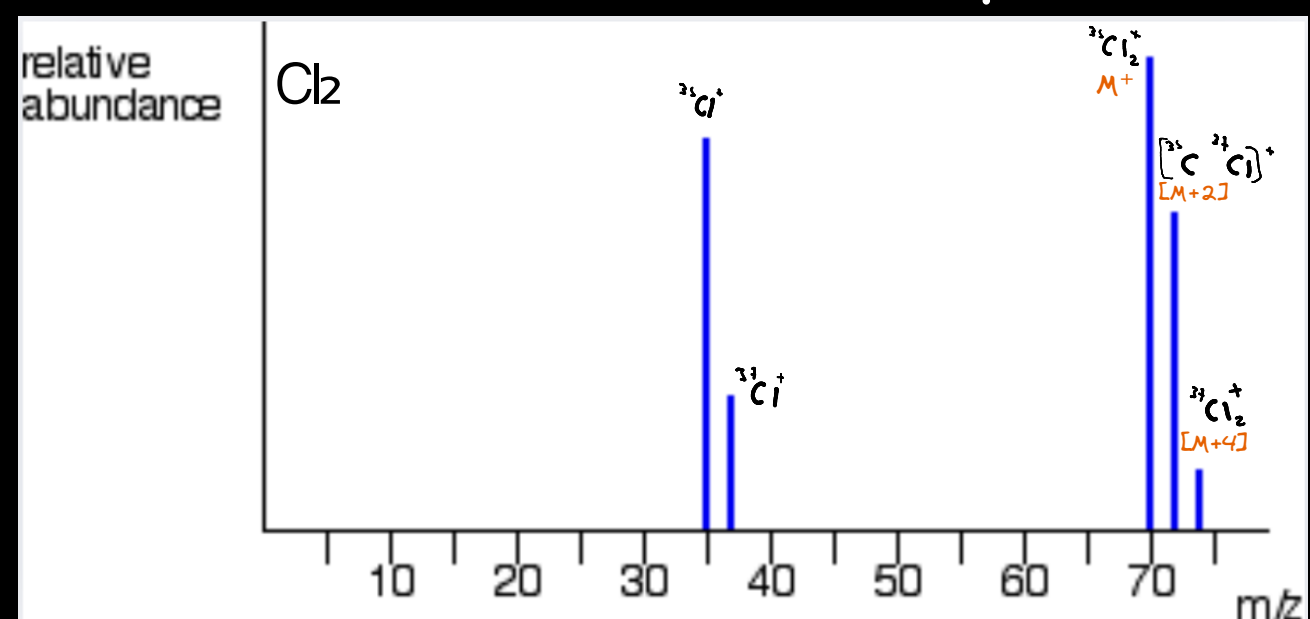
Note:

Sometimes two electrons are removed forming a 2+ charged ions.

Ex Species: $^{10}\text{B}^{2+}$ $m/z = \frac{10}{2} = 5$

Spectra of Molecules

Chlorine



---the ratio of the peak heights for **Chlorine atoms** is 3:1 (as the relative abundance of Cl-35 is 3x greater than that of Cl-37)

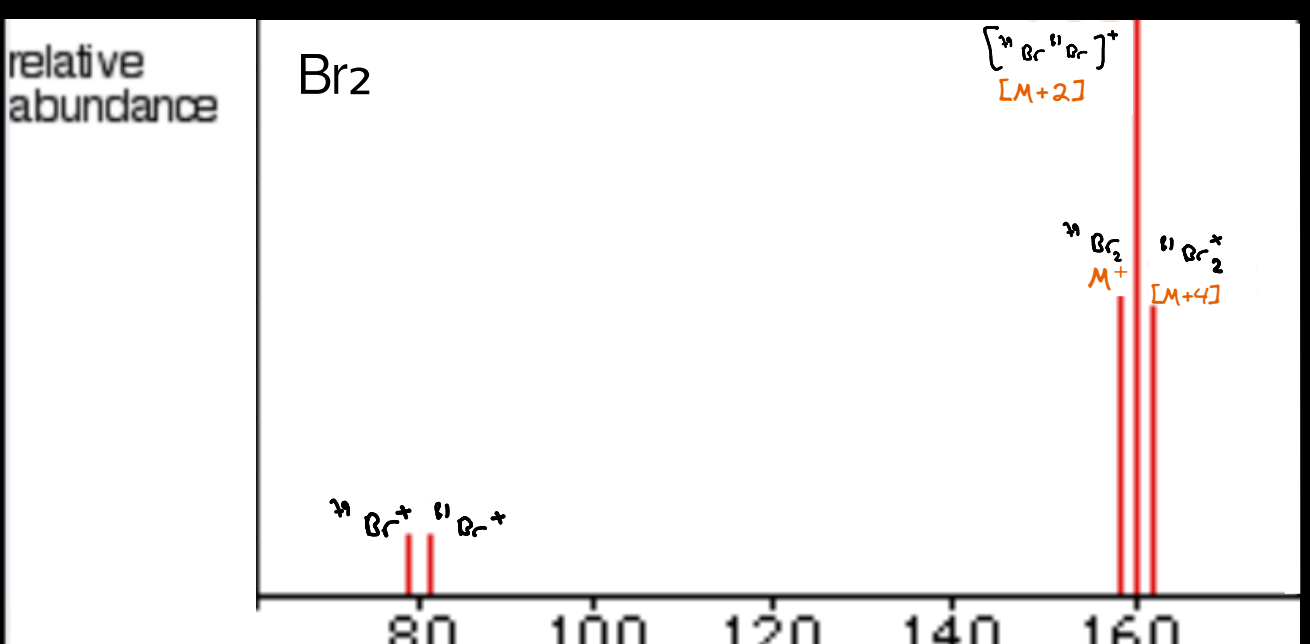
---the ratio of the peak heights for **Chlorine molecules** is 9:4:1

$$\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$$

$$\frac{1}{4} \times \frac{3}{4} \times 2 = \frac{6}{16}$$

$$\frac{1}{4} \times \frac{1}{4} = \frac{1}{16}$$

Bromine



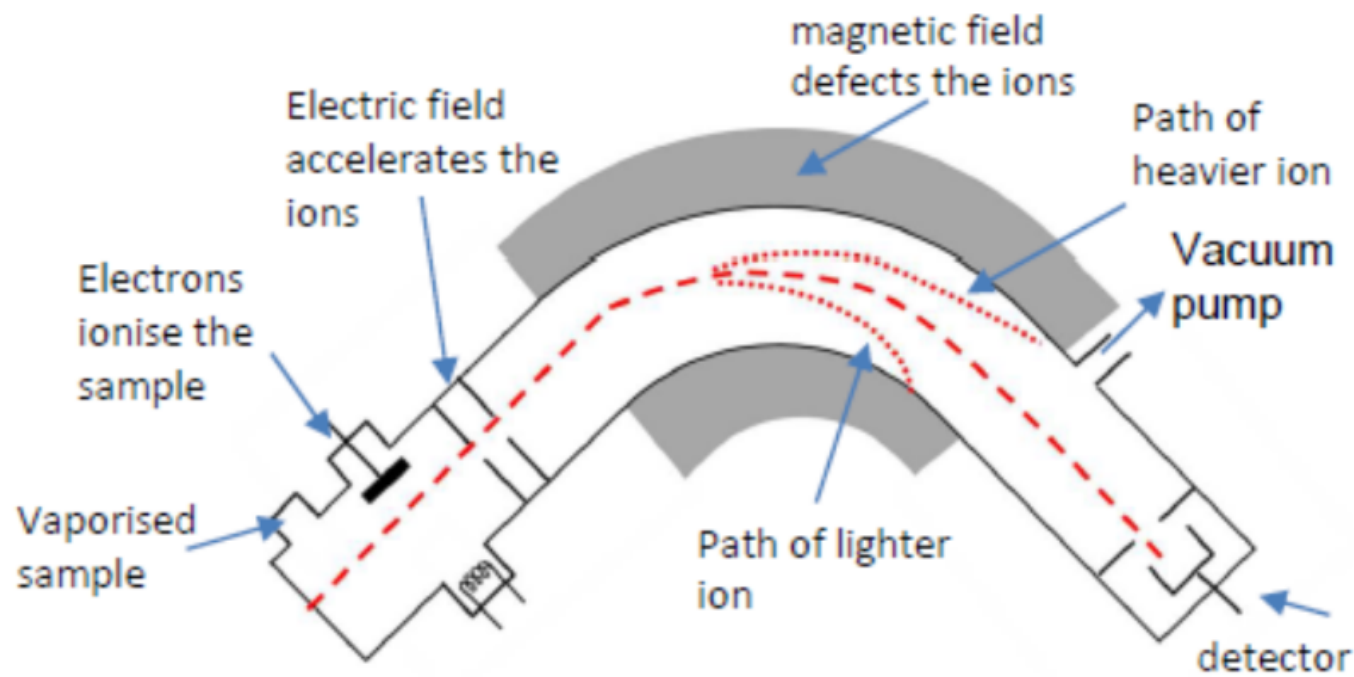
---the ratio of the peak heights for **Bromine atoms** is 1:1 (as the relative abundance of Cl-79 is the same as that of Cl-81)

---the ratio of the peak heights for **Bromine molecules** is 1:2:1

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

$$\frac{1}{2} \times \frac{1}{2} \times 2 = \frac{1}{2}$$

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$



Stages of Mass Spectrometry

Vaporization: The sample must be in its **gaseous state** so that its particles can move through the machine.

Ionization: High-energy electrons collide /bombard the sample, knocking out one or more electrons (mostly only 1) from the atoms or molecules, thereby creating positive ions. equation: $X_2 + e^- \rightarrow X_2^+ + 2e^-$

Acceleration: The ions are **accelerated** by an **electric field**. They then pass through a **velocity selector**, which ensures all ions are moving at the **same velocity**(won't be responsible for difference in deflection).

Deflection: The ions are deflected by a **magnetic field**. The amount of deflection depends on the mass-to-charge ratio (m/z) of the ions.

Detection: The deflected ions pass through a narrow slit and collected on a metallic plate connected to an amplifier. The detector detects how many ions pass through the machine. It shows how many ion have this mass-to-charge ratio(m/z).

----->Only ions with a specific mass-to-charge ratio will be deflected precisely enough to pass through the slit for a given magnetic field strength.

(Larger m/z results in a lower deflection and vice versa)

Greater charge --> Lower m/z (vice versa)

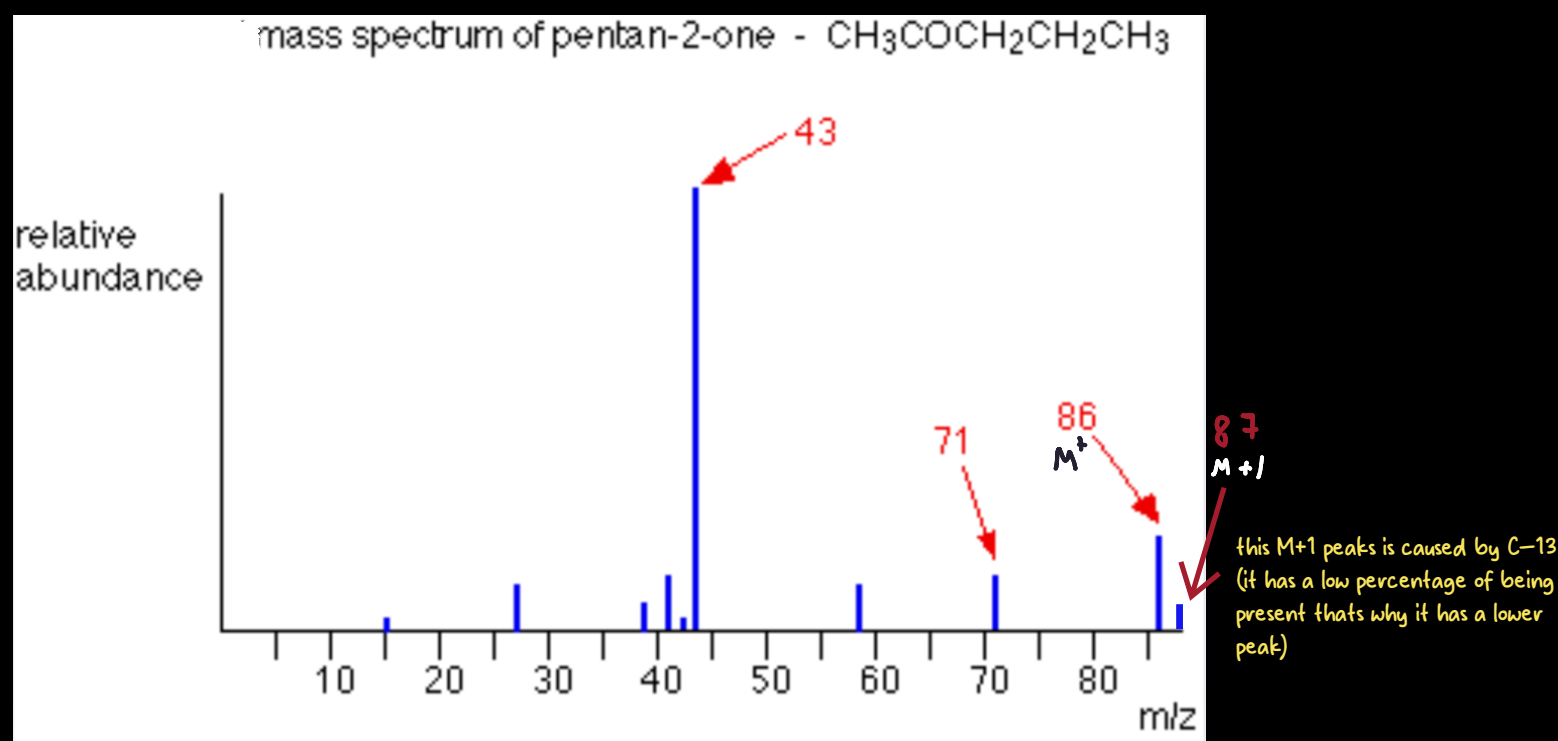
Greater mass --> Greater m/z (vice versa)

----->The magnetic field strength is varied steadily to detect ions with different mass-to-charge ratios.

Why is there a vacuum in the spectrometer ?

1- Ions might collide in air deflecting them from their normal path

2-Air particles can cause additional/false peaks



When a **molecule or compound** is ionized using high-energy electrons not only does it lose an electron (forming a **Molecular/parent ion (M⁺)** which is the Mr of compound) it gets **fragmented** leading to a number of peaks corresponding to fragments formed)

Your Notes

Quantum mechanics

Your Notes

Each principal quantum shell is split into subshells (s, p, d and f) and each it split into smaller regions called orbitals (each orbital holds a max. of 2 electron only)

The electrons fill the subshells starting with the one with least energy (1s) and going up the energy (s < p < d < f)

Principal shell (n)	Subshells	Orbitals	Electrons(e-)
1	1s	1	2
2	2s, 2p	1 + 3 = 4	2 + 6 = 8
3	3s, 3p, 3d	1 + 3 + 5 = 9	2 + 6 + 10 = 18
n ≥ 4	4s, 4p, 4d, 4f	1 + 3 + 5 + 7 = 16	2 + 6 + 10 + 14 = 32

Subshells

s - subshell

Orbitals

Shape: spherical

Number: 1 (2e⁻)

s-Block

Elements: group 1, 2, H and He

Why?: Their highest energy

electron is in the s subshell

p - subshell

Orbitals

Shape: dumbbell

Number: 3 (6e⁻)

p-Block

Elements: group 3-8

Why?: Their highest energy

electron is in the p subshell

Orbital: the **region** of space within a nucleus where electrons are **most likely** to be found (90%–95%)

2-Short hand

Rules:

1-strat with subshell that has the least energy

2-Replace the inner electrons that match a

noble gas configuration with the symbol of that noble gas in brackets

Example:

Na (Z = 11):

Full: 1s² 2s² 2p⁶ 3s¹

Nearest noble gas before Na: Neon (Ne) — 1s² 2s² 2p⁶

Shorthand: [Ne] 3s¹

d - subshell

Orbitals

Number: 5 (10e⁻)

d-block

Elements: between groups 2, 3

Why?: Their highest energy

electron is in the d subshell

f - subshell

Orbitals

Number: 7 (14e⁻)

f-block

Elements: lanthanides and actinides

Why?: Their highest energy

electron is in the f subshell

S P D F Blocks

s-block		p-block
1s		1s
2s		2s
3s		3s
4s		4s
5s		5s
6s		6s
7s		7s
	d-block	
	3d	
	4d	
	5d	
	6d	
	f-block	
	4f	
	5f	

Electronic configuration of atoms

1-Box form

Rules:

1-strat with subshell that has the least energy

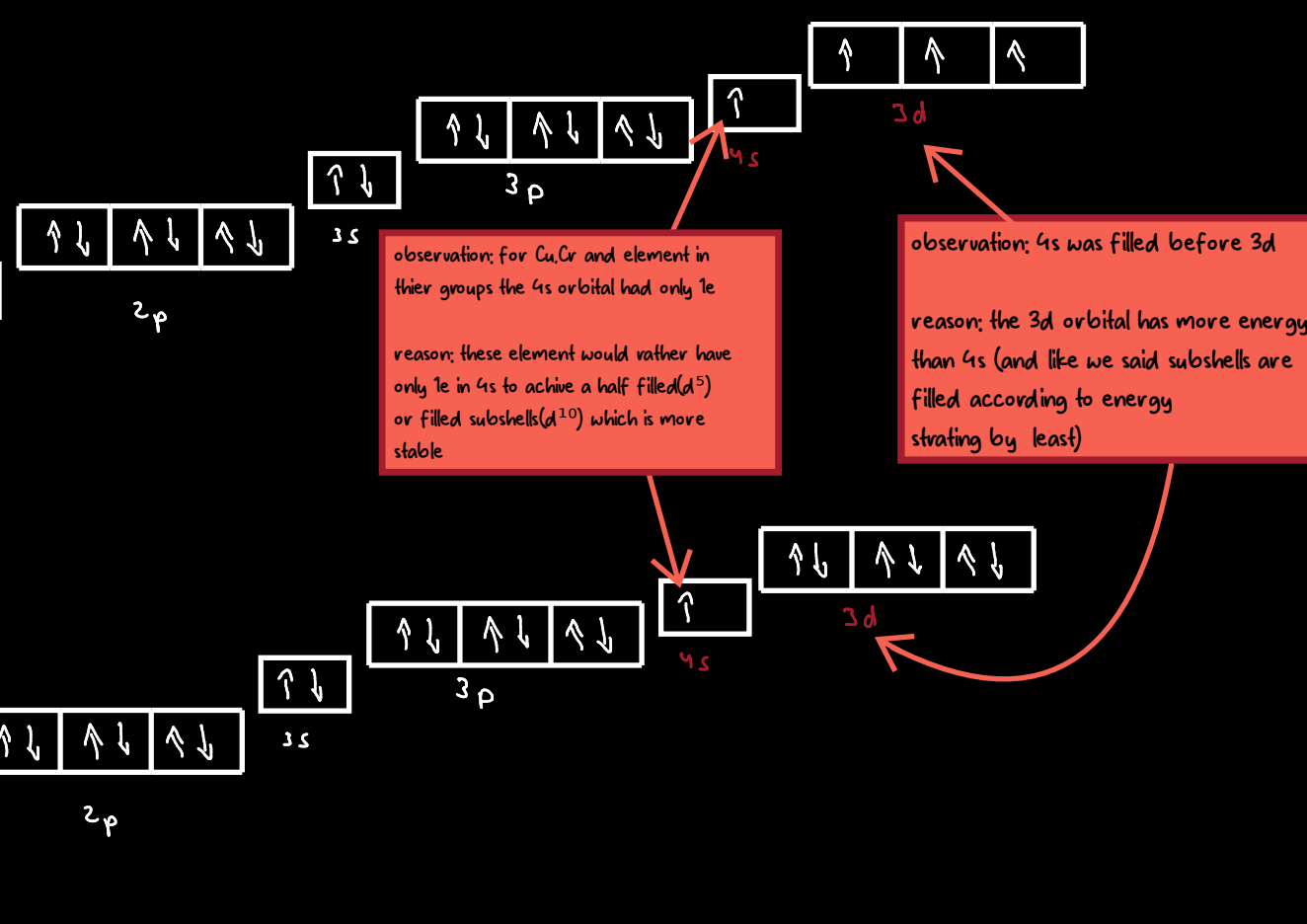
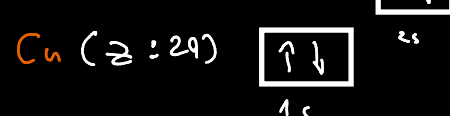
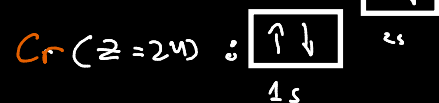
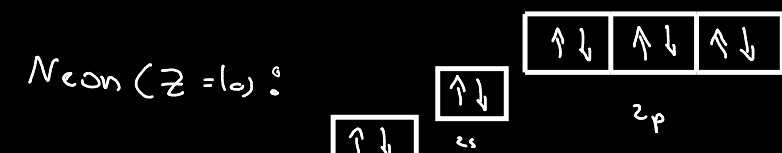
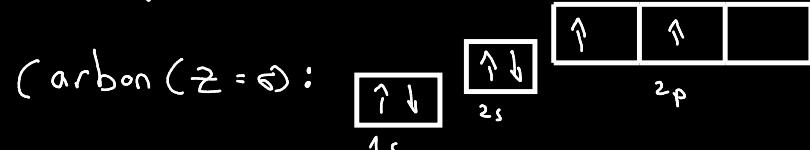
2-e⁻ will occupy the orbital singly before pairing

takes place

3-e⁻ in same orbital has opposite spins

before pairing takes place

Examples:

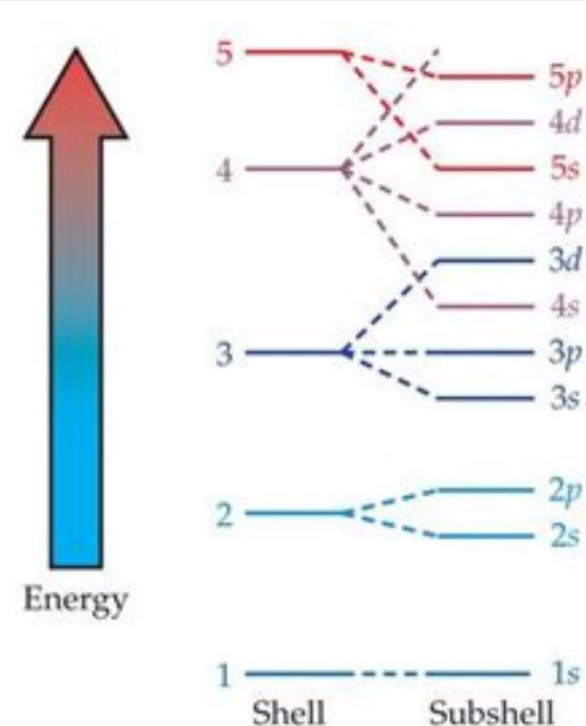


3— Full configuration

Rules:

1—strat with subshell that has the least energy (1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p.)

#	Element	Symbol	Electronic Configuration
1	Hydrogen	H	$1s^1$
2	Helium	He	$1s^2$
3	Lithium	Li	$1s^2 2s^1$
4	Beryllium	Be	$1s^2 2s^2$
5	Boron	B	$1s^2 2s^2 2p^1$
6	Carbon	C	$1s^2 2s^2 2p^2$
7	Nitrogen	N	$1s^2 2s^2 2p^3$
8	Oxygen	O	$1s^2 2s^2 2p^4$
9	Fluorine	Fl	$1s^2 2s^2 2p^5$
10	Neon	Ne	$1s^2 2s^2 2p^6$
11	Sodium	Na	$1s^2 2s^2 2p^6 3s^1$
12	Magnesium	Mg	$1s^2 2s^2 2p^6 3s^2$
13	Aluminium	Al	$1s^2 2s^2 2p^6 3s^2 3p^1$
14	Silicon	Si	$1s^2 2s^2 2p^6 3s^2 3p^2$
15	Phosphorous	P	$1s^2 2s^2 2p^6 3s^2 3p^3$
16	Sulphur	S	$1s^2 2s^2 2p^6 3s^2 3p^4$
17	Chlorine	Cl	$1s^2 2s^2 2p^6 3s^2 3p^5$
18	Argon	Ar	$1s^2 2s^2 2p^6 3s^2 3p^6$
19	Potassium	K	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
20	Calcium	Ca	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
21	Scandium	Sc	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$
22	Titanium	Ti	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$
23	Vanadium	V	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$
24	Chromium*	Cr	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$
25	Manganese	Mn	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
26	Iron	Fe	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$
27	Cobalt	Co	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
28	Nickel	Ni	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$
29	Copper*	Cu	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$
30	Zinc	Zn	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$
31	Gallium	Ga	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^1$
32	Germanium	Ge	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$
33	Arsenic	As	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$
34	Selenium	Se	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$
35	Bromine	Br	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$
36	Krypton	Kr	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$



observation: 4s was filled before 3d

reason: the 3d orbital has more energy than 4s (and like we said subshells are filled according to energy strating by smallest)

Electronic configuration of ions

Rules:

1—remove the e^- from the obital with highest energy

2—the electron are removed from 4s before 3d

Your Notes

Atomic radius

Down a group: the atomic radius **increases**:

Addition of a New Shell: Each element adds a new principal energy level (e.g., $n=2$ to $n=3$), placing the valence electrons farther from the nucleus.

Increased Shielding: The inner core of electrons shields the valence electrons from the full attractive force of the nucleus.

Increased Energy/repulsion: More electrons in the outer shell lead to greater electron–electron repulsion, which pushes the electron cloud outward.

Weaker Effective Nuclear Pull: The combined effect of greater distance and stronger shielding means the valence electrons experience a weaker force (even though the nuclear charge increases).

Across a period (left to right), the atomic radius **decreases**.

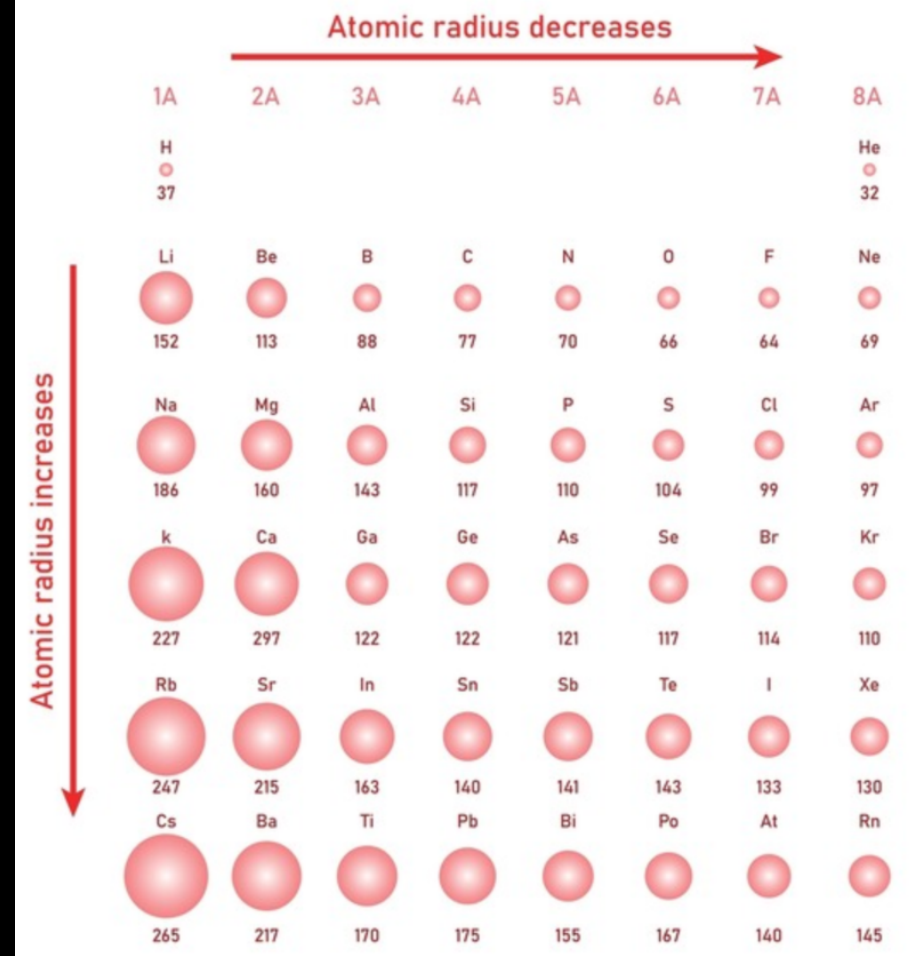
Increasing Nuclear Charge: The number of protons in the nucleus increases with each element.

Constant Shielding: Electrons are added to the same principal energy level, so the shielding by inner electrons remains constant.

Stronger Nuclear Pull: Increasing attractive forces between the nucleus and the electrons

More tightly held

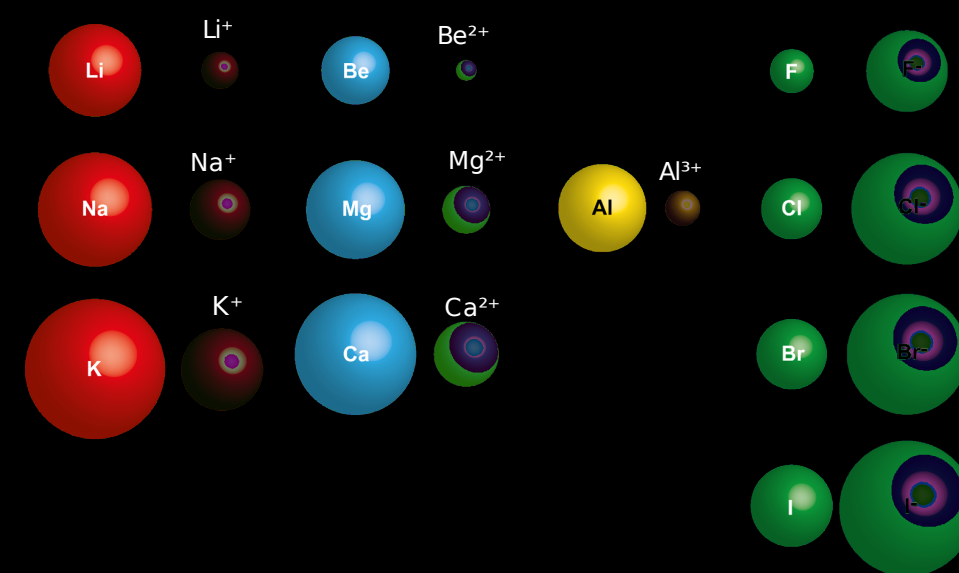
Note: this is only partially offset by the increased electron (–electron) repulsion as the number of electrons in the (outer) shell increases



Atomic radius: the distance between nucleus and boundary (outermost) of the electron cloud

its a measure of the size of an atom

Shielding: an effect of inner electrons which reduces the pull of the nucleus on the electrons in the outer shell of an atom



Ionic radius

Cations are always **smaller** than their parent atoms.

Loss of an Electron Shell: When a cation forms, the entire outer valence shell of electrons is often lost.

Increased Proton–to–Electron Ratio: The atom loses electrons but retains the same number of protons, leading to a greater nuclear charge per remaining electron.

Reduced Electron Repulsion: Fewer electrons mean less electron–electron repulsion, allowing the electron cloud to be pulled inward.

More tightly held

Anions are always **larger** than their parent atoms.

Gain of Electrons: The atom gains extra electrons into its existing **valence shell**.

Decreased Proton–to–Electron Ratio: The nuclear charge remains the same, but it is now spread over a larger number of electrons, reducing the nuclear charge per electron

Increased Electron Repulsion: The additional electrons increase the repulsive forces within the electron cloud, pushing electrons further apart.

less tightly held

Ionization energy

Ionisation energy (IE): The amount of energy needed to remove one mole of electrons from one mole of gaseous **atoms** (endothermic) **unit: kJ/mol**

1st Ionization Energy (IE₁): The amount of energy needed to remove one mole of the most loosely held electrons from one mole of gaseous atoms to form one mole of +1 gaseous cations.

General Equation: $X(g) \rightarrow X^+(g) + e^-$

Example: $Na(g) \rightarrow Na^+(g) + e^-$

2nd Ionization Energy (IE₂): The amount of energy needed to remove one mole of electrons from one mole of +1 gaseous cations to form one mole of +2 gaseous cations.

General Equation: $X^+(g) \rightarrow X^{2+}(g) + e^-$

Example: $Na^+(g) \rightarrow Na^{2+}(g) + e^-$

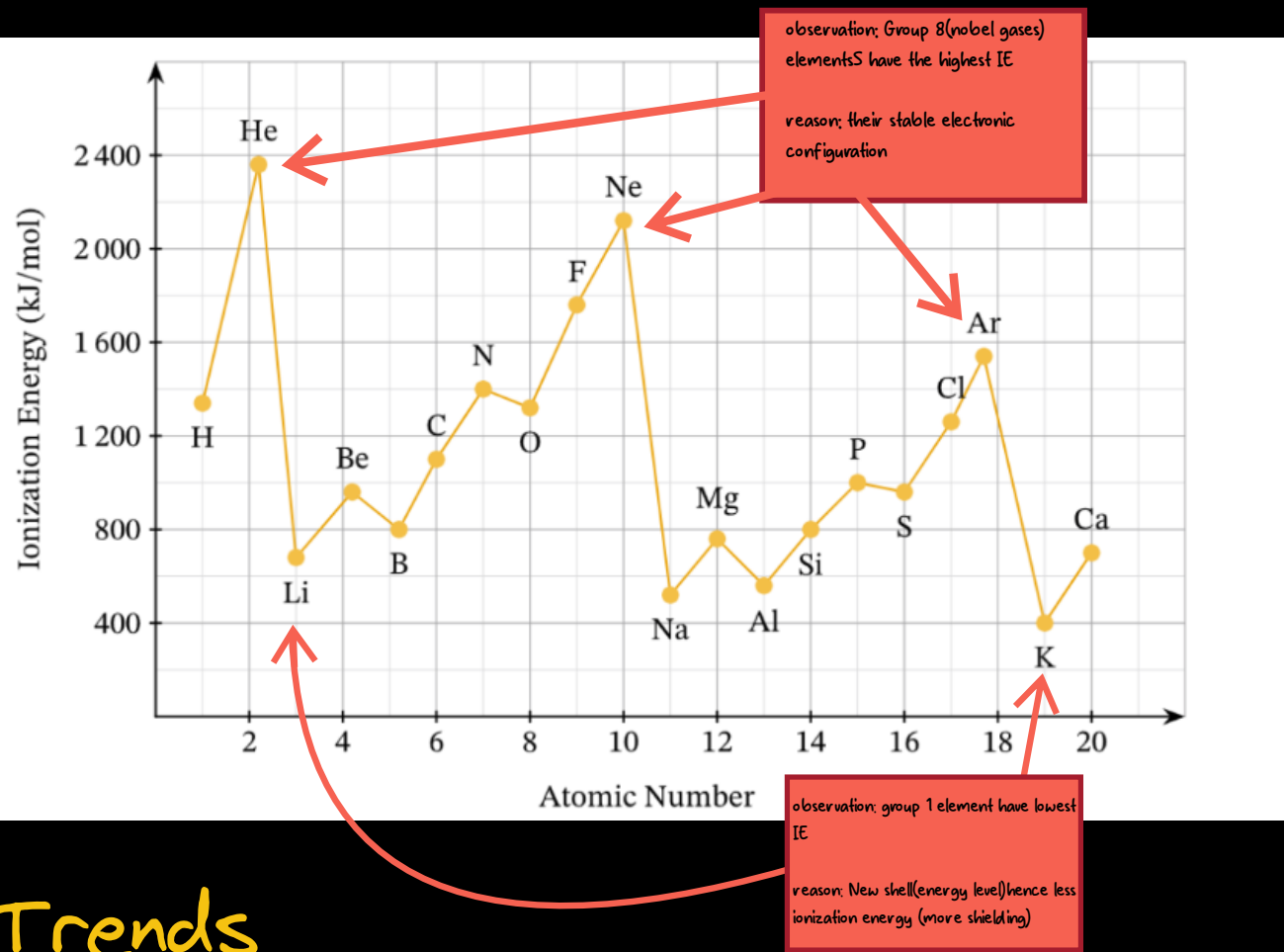
3rd Ionization Energy (IE₃): The amount of energy needed to remove one mole of electrons from one mole of +2 gaseous cations to form one mole of +3 gaseous cations.

General Equation: $X^{2+}(g) \rightarrow X^{3+}(g) + e^-$

Example: $Mg^{2+}(g) \rightarrow Mg^{3+}(g) + e^-$

Your Notes

Ionization energy



Trends

- IE Increases Across a Period**
- Shielding is constant (same shells)
 - Number of protons increases (greater nuclear charge)
 - Stronger attraction between the nucleus and electrons
 - Smaller atomic radius
- IE Decreases Down a Group**
- New electron shell added
 - Outer electron in further to the nucleus (greater atomic radius)
 - Increased shielding from inner electrons (electron-electron repulsion)
 - Increased Energy (level)
 - Result: The nuclear pull on the outermost electron weakens.

Exception: Boron < Beryllium (Al < Mg)

B: Electron removed from a higher energy 2p orbital.

Be: Electron removed from a stable, lower energy 2s orbital.

The 2p electron is also shielded by the 2s subshell and on a higher energy evel, making it easier to remove.

Exception: Oxygen < Nitrogen (Sulfur < Phosphorus)

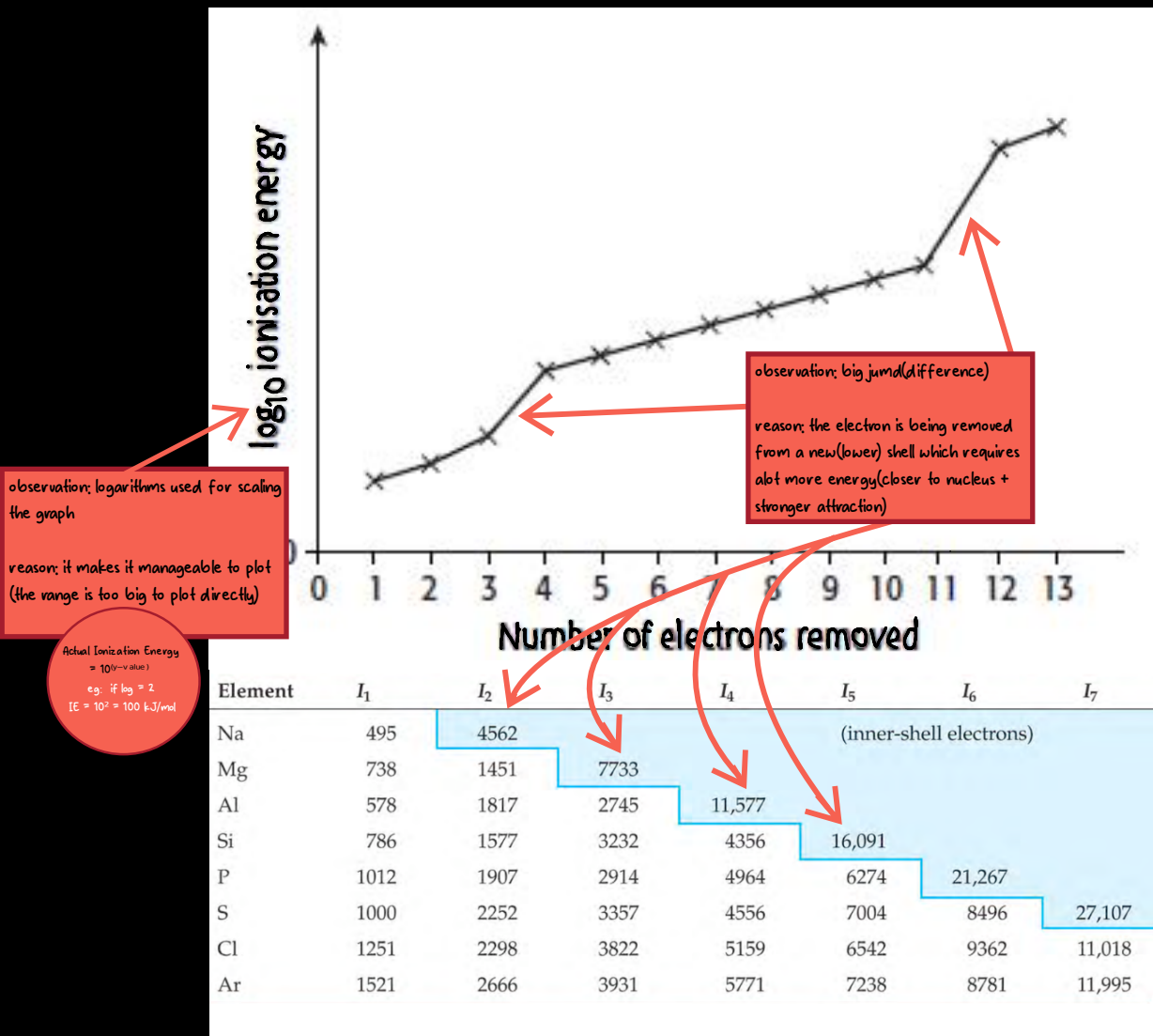
both: Removed from p shell

O/S: Paired electron removed from a 2p orbital with 2 electrons (pair).

N/P: Unpaired electron removed from a stable, half-filled 2p subshell.

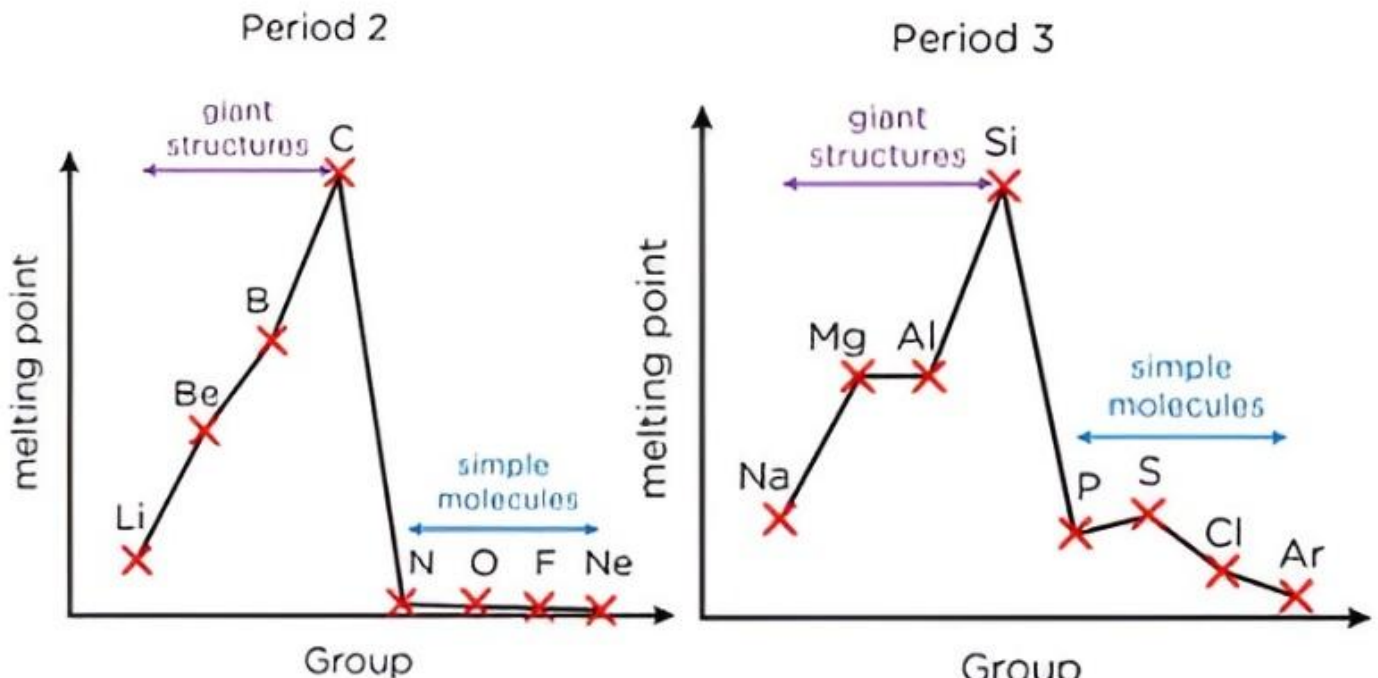
Spin-pair repulsion in the paired orbital makes the electron easier to remove (reduces ionization energy).

Successive ionization energies



- There would be a general increase
- 1—electrons are removed (less repulsion)
 - 2—radius decreases
 - 3—greater electrostatic attraction

Melting and boiling points



Period 2	Li	Be	B	C	N	O	F
M.P/B.P	High	High	High	High	Low	Low	Low
Conductivity	good	good	bad	bad	bad	bad	bad
bonding type	Metalic	Metalic	Covelant	Covelant	Covelant	Covelant	Covelant
structure	Giant	Giant	Giant	Giant	Simple molecular	Simple molecular	Simple molecular

Period 3	Na	Mg	Al	Si	P	S	Cl
M.P/B.P	High	High	High	High	Low	Low	Low
Conductivity	good	good	good	bad	bad	bad	bad
bonding type	Metalic	Metalic	Metalic	Covelant	Covelant	Covelant	Covelant
structure	Giant	Giant	Giant	Giant	Simple molecular	Simple molecular	Simple molecular

- Across period**
- Na—Al (Li—Be):**
- 1—melting/boiling points: increase as there are strong metallic bond that get stronger (harder to overcome) with more delocalized electron
 - 2—Conductivity: Good and increases as there are more delocalized electron
- Si (B—C):**
- 1—Melting/boiling points: high (highest at C/Si) as they are held by strong covelant bonds forming a Giant structure (requires a lot of energy to overcome forces)
 - 2—Conductivity: bad as there are no free electrons
- P—Cl (N—F):**
- 1—Melting/boiling points: low (lowest at Cl/F) as molecules are held by weak forces (London force) forming a simple molecular structure that are easy and require little energy to overcome. Among the common molecular elements in Period 3, Sulfur (S₈) has the strongest intermolecular forces due to its larger molecule.
 - 2—Conductivity: bad as there are no free electrons

Your Notes